Technical Skills

Programming Languages: Python, Java, C, C++, JavaScript, HTML5, CSS3, SQL
General: Fine-tuning, Prompt Engineering, Agile, Data Scraping, Shell Scripting, Containerization, Multithreading, CI/CD, Automated Testing, Cloud Deployment, Network Protocols
Frameworks/Tools: Linux, AWS, GCP, Docker, ROS, Pytorch, Tensorflow, Django, React, Node.js, Express, JUnit

Experience

Software Engineer March 2021 – July 2022
EarthSense, Inc. *Champaign, IL*

- Designed an algorithm to estimate plant height using noisy LiDAR data collected by our phenotyping robot, cutting error by >50% across datasets over the previous algorithm and beating accuracy of manual measurements.
- Revamped robot autonomy by designing a crash detection algorithm fusing LiDAR and odometry data collected from various environments, reducing overall false positives and negatives by 75%.
- Greatly decreased need for manual waypoint recording by implementing automatic waypoint generation on turns, saving customers several minutes per data collection and improving robot autonomy.
- Introduced a culture of documentation in our workflows for libraries and testing methodologies, which I later used to onboard interns for a summer.
- Extensively tested robot system functionality, standardized testing methodologies and created utility shell scripts, helping catch several system bugs before deployment to customers.

Software Engineer March 2021 – July 2022
CONDENSED - EarthSense, Inc. *Champaign, IL*

- Designed an algorithm to estimate plant height using noisy LiDAR data, cutting error by >50% across datasets over the previous algorithm and beating accuracy of manual measurements.
- Revamped robot autonomy by designing and finetuning a crash detection algorithm fusing LiDAR and odometry data collected from various sensors, reducing overall errors in crash detection by 75%.
- Implemented automatic waypoint generation for robot navigation, saving customers hours during setup time.
- Standardized testing methodologies and created shell scripts to catch system bugs before deployment.

Teaching and Research Assistant - AI For Redistricting January 2025 – May 2025
University of San Francisco *San Francisco, CA*

Projects

Template Month Year

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Duets AI August - December 2025

- Led a 3-person industry-sponsored team shipping LLM-backed features for a language-learning web app, using Python, Django, Azure and OpenAI.
- Cut company-wide LLM spend by 80% by switching models, prompt engineering, and backend changes.
- Lowered server and storage costs by 30% by reworking frontend and backend logic, and migrating tens of thousands of hot data entries from cold storage to Supabase.
- Built meeting transcript-based exercise generation; added checks, retries, and fallbacks to make it 99.9% reliable.
- Coordinated scope and delivery with sponsors and two student teams; enforced issue/PR tracking and review standards to keep releases moving efficiently.

Highly Available Cloud Application Deployment October 2024

- Architected and deployed a highly available, secure application on AWS using ECR, EC2, ALB, VPC, and ASGs.
- Configured an ALB to distribute traffic across EC2 instances in private subnets across multiple AZ's with launch templates and auto-scaling groups, enhancing scalability, security and availability.
- Improved operational security by launching bastion hosts across availability zones using ASGs for highly available, secure SSH access to EC2 instances.
- Set up TLS encryption using ACM and Route 53 for domain management and HTTPS-encrypted communication.

LLMs To Detect Chart Misinformation August 2024

- Finetuned a multimodal LLM (LLaVA) to detect visual misinformation in bar charts with LoRA and 2.3k examples.
- Created a custom finetuning script and explored parameters to prevent overfitting and minimize GPU loads.
- Resulted in a model that can identify visually manipulated bar charts 90% of the time.

Syllabus Generator May 2024

- Used low-cost LLMs and crewAI to coordinate the automatic generation of a syllabus using basic course info.
- Implemented prompt engineering techniques to optimize syllabus generation, significantly improving output.
- Developed a microservice using FastAPI that talks to large LLMs like ChatGPT for automated syllabus generation.

Craigslist Apartment Scraper July 2023

- A Python web-scraping script that enables a user to browse hundreds of ads for rental accomodation on Craigslist in a fraction of the time it would take to do so using the website.
- Scraped results are output to HTML files in a readable manner according to a list of user-specified constraints on commute time, rent, number of bedrooms, location, shared versus whole apartments, and more.
- Allows a user to easily see commute times to a specified location by foot, bike, and car. This data is queried from OpenStreetMap and can be used to sort listings according to travel time buckets by a chosen mode of transport.

Mindset November 2023

- A wellness web-app with a task management system to help users track and quit addictions, track nutrition, sleep quality and focus on work.
- Group project where we used Node.js with Express to set up the server, Passport.js and Google OAuth2 for authentication, and Mongoose to store user data.

Reddit Product Scraper March 2021

- Solo project to find specific products being re-sold on Reddit subreddits to get them at a discount.
- Leveraged the Reddit API to scrape thousands of posts per minute to find relevant products and deliver them to the user in a readable fashion.
- Personally saved $150 on $500 worth of purchases.

Blackjack Agent November 2023

- Developed an agent to play blackjack against humans using Q-learning as a reinforcement learning algorithm.
- Simulated an environment to teach an agent to optimize decision making by balancing exploration and exploitation.
- Used matplotlib and the learned policy of the agent to visualize agent policy.

Fashion Image Classifier December 2023

- Used keras and tensorflow to adapt ViT (Vision Transformer) to classify fashion articles by training on the fashion_mnist dataset.
- Achieved a train accuracy of 86.88% after 4 epochs of training on 4000 images.